

IMAGE SUPER-RESOLUTION MODELS PERFORMANCE BENCHMARK REPORT

Empirical Benchmark: Bicubic Baseline vs. Deep Learning Models (Compact RTL & Software Architectures)

BENCHMARKED ARCHITECTURES & EVALUATED MODELS:

- 1. Bicubic (Baseline): Standard bicubic polynomial interpolation algorithm (non-parametric baseline).
- 2. **Compact SRCNN RTL: Hardware-compact architecture 1 -> 16 -> 8 -> 1 (1,649 parameters, INT8 Q7).**
- 3. **SRCNN Original: Original CNN baseline architecture 1 -> 64 -> 32 -> 1 (8,129 parameters, FP32).**
- 4. ESPCN: Efficient Sub-Pixel Convolutional Neural Network (PixelShuffle upscale in LR space).
- 5. FSRCNN: Fast Super-Resolution CNN with feature dimension shrinking and expanding layers.
- 6. VDSR: Very Deep Super-Resolution network with 20 convolutional layers and residual learning.
- 7. EDSR: Enhanced Deep Residual Networks (8 ResBlocks, 64 deep feature channels).
- 8. SRGAN: Super-Resolution Generative Adversarial Network optimized for perceptual quality.

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I. DATASET: SUB_NIH (NIH CHESTX-RAY14)

1. Magnification Factor: Scale 2x

Metric	Bicubic (Baseline)	Compact SRCNN RTL	SRCNN Original	ESPCN	FSRCNN	VDSR	EDSR	SRGAN
PSNR (dB) ↑	40.0682 dB	39.3011 dB	39.6171 dB	37.2428 dB	31.9671 dB	40.3761 dB	40.3739 dB	37.5680 dB
SSIM ↑	0.9753	0.9668	0.9737	0.9610	0.9254	0.9777	0.9783	0.9636
MS-SSIM ↑	0.9976	0.9966	0.9967	0.9941	0.9724	0.9978	0.9980	0.9947
LPIPS ↓	0.0956	0.0583	0.1653	0.2581	0.4316	0.1140	0.0863	0.2118
NIQE ↓	9.9243	7.6964	6.9968	7.8967	10.3887	6.7623	6.2820	6.6020
EPI ↑	0.3274	0.3039	0.4451	0.1479	-0.0508	0.4693	0.4076	0.2412
MSE ↓	8.2065	9.0190	9.1015	15.1724	45.8111	7.6720	7.8904	14.7761
RMSE ↓	2.7030	2.8844	2.8460	3.6969	6.5953	2.6124	2.6330	3.6104
Latency (ms) ↓	15.33 ms	12.73 ms	44.61 ms	4.80 ms	8.11 ms	336.82 ms	111.59 ms	318.63 ms
Throughput (FPS) ↑	65.2 FPS	78.6 FPS	22.4 FPS	208.5 FPS	123.3 FPS	3.0 FPS	9.0 FPS	3.1 FPS

* Notes: (↑) Higher is better | (↓) Lower is better | Bold: Optimal metric value across benchmarked models
Bicubic (Baseline): Standard interpolation baseline | Compact SRCNN RTL (1-16-8-1, 1,649 params, INT8 Q7) | SRCNN Original (1-64-32-1, 8,129 params, FP32)

I. DATASET: SUB_NIH (NIH CHESTX-RAY14)

2. Magnification Factor: Scale 3x

Metric	Bicubic (Baseline)	Compact SRCNN RTL	SRCNN Original	ESPCN	FSRCNN	VDSR	EDSR	SRGAN
PSNR (dB) ↑	37.5998 dB	38.4493 dB	39.2991 dB	36.2036 dB	18.9758 dB	39.2127 dB	39.4904 dB	37.1230 dB
SSIM ↑	0.9620	0.9538	0.9643	0.9423	0.6448	0.9647	0.9658	0.9489
MS-SSIM ↑	0.9936	0.9930	0.9941	0.9871	0.8146	0.9941	0.9946	0.9900
LPIPS ↓	0.2076	0.1214	0.2478	0.3216	1.0178	0.2363	0.2079	0.2945
NIQE ↓	9.9727	7.8957	7.7859	8.7711	27.3127	7.8913	7.2587	8.2229
EPI ↑	0.1663	0.1064	0.2773	0.0457	0.0043	0.2546	0.2249	0.1132
MSE ↓	14.9982	11.0340	9.4633	18.7101	852.7464	9.7603	9.0453	15.3459
RMSE ↓	3.6182	3.1787	2.9142	4.1291	28.9475	2.9520	2.8508	3.7284
Latency (ms) ↓	13.67 ms	12.87 ms	45.47 ms	2.70 ms	4.22 ms	395.91 ms	64.26 ms	241.86 ms
Throughput (FPS) ↑	73.2 FPS	77.7 FPS	22.0 FPS	370.5 FPS	236.8 FPS	2.5 FPS	15.6 FPS	4.1 FPS

* Notes: (↑) Higher is better | (↓) Lower is better | Bold: Optimal metric value across benchmarked models
Bicubic (Baseline): Standard interpolation baseline | Compact SRCNN RTL (1-16-8-1, 1,649 params, INT8 Q7) | SRCNN Original (1-64-32-1, 8,129 params, FP32)

I. DATASET: SUB_NIH (NIH CHESTX-RAY14)

3. Magnification Factor: Scale 4x

Metric	Bicubic (Baseline)	Compact SRCNN RTL	SRCNN Original	ESPCN	FSRCNN	VDSR	EDSR	SRGAN
PSNR (dB) ↑	36.1873 dB	35.8637 dB	35.5253 dB	21.4287 dB	15.5623 dB	36.3261 dB	36.4724 dB	35.1208 dB
SSIM ↑	0.9495	0.9413	0.9501	0.6679	0.6252	0.9529	0.9530	0.9406
MS-SSIM ↑	0.9900	0.9893	0.9891	0.8780	0.8155	0.9906	0.9911	0.9859
LPIPS ↓	0.2865	0.1773	0.3408	0.6851	0.9235	0.3185	0.3000	0.3489
NIQE ↓	9.9859	8.4658	9.3520	16.9979	272.5003	8.8045	8.2138	8.9861
EPI ↑	0.0923	0.0309	0.2054	-0.0005	-0.0070	0.1309	0.1015	0.0835
MSE ↓	20.6699	21.5248	24.2238	484.6246	1874.2795	20.1630	19.4813	25.6018
RMSE ↓	4.2471	4.3670	4.5899	21.8243	42.8994	4.1883	4.1170	4.7600
Latency (ms) ↓	12.71 ms	12.73 ms	23.21 ms	1.74 ms	2.56 ms	368.23 ms	39.42 ms	195.89 ms
Throughput (FPS) ↑	78.7 FPS	78.6 FPS	43.1 FPS	574.6 FPS	390.1 FPS	2.7 FPS	25.4 FPS	5.1 FPS

* Notes: (↑) Higher is better | (↓) Lower is better | Bold: Optimal metric value across benchmarked models
Bicubic (Baseline): Standard interpolation baseline | Compact SRCNN RTL (1-16-8-1, 1,649 params, INT8 Q7) | SRCNN Original (1-64-32-1, 8,129 params, FP32)

II. DATASET: SUB_CHEST (CLINICAL CHEST X-RAY)

1. Magnification Factor: Scale 2x

Metric	Bicubic (Baseline)	Compact SRCNN RTL	SRCNN Original	ESPCN	FSRCNN	VDSR	EDSR	SRGAN
PSNR (dB) ↑	40.9900 dB	39.2359 dB	39.5278 dB	37.4337 dB	32.3510 dB	39.9936 dB	40.3111 dB	38.1845 dB
SSIM ↑	0.9597	0.9308	0.9332	0.9078	0.8558	0.9432	0.9468	0.9119
MS-SSIM ↑	0.9960	0.9947	0.9921	0.9862	0.9542	0.9950	0.9957	0.9878
LPIPS ↓	0.1427	0.1412	0.2778	0.3876	0.5941	0.2027	0.1791	0.3399
NIQE ↓	9.9425	7.8182	6.9180	8.3889	9.7426	5.9437	5.6151	6.6622
EPI ↑	0.3991	0.2491	0.3511	0.0867	0.0002	0.3707	0.3718	0.2173
MSE ↓	5.3800	8.0486	7.6855	12.2547	38.8168	6.8945	6.4393	10.4249
RMSE ↓	2.2965	2.8108	2.7325	3.4636	6.1908	2.5895	2.4997	3.1849
Latency (ms) ↓	15.39 ms	492.30 ms	124.29 ms	13.02 ms	20.46 ms	861.29 ms	308.03 ms	799.78 ms
Throughput (FPS) ↑	65.0 FPS	2.0 FPS	8.0 FPS	76.8 FPS	48.9 FPS	1.2 FPS	3.2 FPS	1.3 FPS

* Notes: (↑) Higher is better | (↓) Lower is better | Bold: Optimal metric value across benchmarked models
Bicubic (Baseline): Standard interpolation baseline | Compact SRCNN RTL (1-16-8-1, 1,649 params, INT8 Q7) | SRCNN Original (1-64-32-1, 8,129 params, FP32)

II. DATASET: SUB_CHEST (CLINICAL CHEST X-RAY)

2. Magnification Factor: Scale 3x

Metric	Bicubic (Baseline)	Compact SRCNN RTL	SRCNN Original	ESPCN	FSRCNN	VDSR	EDSR	SRGAN
PSNR (dB) ↑	38.4502 dB	37.5675 dB	38.1119 dB	35.8167 dB	19.4360 dB	38.0783 dB	38.3483 dB	36.8212 dB
SSIM ↑	0.9338	0.8993	0.9115	0.8779	0.6049	0.9132	0.9162	0.8836
MS-SSIM ↑	0.9885	0.9858	0.9859	0.9745	0.8140	0.9863	0.9874	0.9785
LPIPS ↓	0.2876	0.2417	0.4035	0.4873	1.0019	0.3675	0.3308	0.4806
NIQE ↓	9.9710	7.5179	7.8520	8.5688	21.9968	7.6123	7.1297	8.7333
EPI ↑	0.2136	0.0751	0.1927	0.0256	0.0004	0.1811	0.1767	0.1077
MSE ↓	9.5621	11.8406	10.5679	17.6864	748.9688	10.6449	10.0377	14.1808
RMSE ↓	3.0694	3.4070	3.2095	4.1663	27.2897	3.2216	3.1255	3.7202
Latency (ms) ↓	13.73 ms	444.88 ms	124.54 ms	7.62 ms	11.01 ms	934.05 ms	172.38 ms	558.56 ms
Throughput (FPS) ↑	72.8 FPS	2.2 FPS	8.0 FPS	131.2 FPS	90.8 FPS	1.1 FPS	5.8 FPS	1.8 FPS

* Notes: (↑) Higher is better | (↓) Lower is better | Bold: Optimal metric value across benchmarked models
Bicubic (Baseline): Standard interpolation baseline | Compact SRCNN RTL (1-16-8-1, 1,649 params, INT8 Q7) | SRCNN Original (1-64-32-1, 8,129 params, FP32)

II. DATASET: SUB_CHEST (CLINICAL CHEST X-RAY)

3. Magnification Factor: Scale 4x

Metric	Bicubic (Baseline)	Compact SRCNN RTL	SRCNN Original	ESPCN	FSRCNN	VDSR	EDSR	SRGAN
PSNR (dB) ↑	36.5994 dB	36.5373 dB	36.2227 dB	21.9458 dB	16.0441 dB	36.9682 dB	37.1728 dB	36.0685 dB
SSIM ↑	0.9114	0.8779	0.8898	0.6314	0.5881	0.8937	0.8945	0.8736
MS-SSIM ↑	0.9823	0.9795	0.9781	0.8779	0.8088	0.9802	0.9811	0.9724
LPIPS ↓	0.3955	0.3138	0.5324	0.7220	0.9233	0.4913	0.4775	0.5385
NIQE ↓	9.9821	8.1220	9.4849	14.7369	67.3225	8.7126	8.3824	9.2006
EPI ↑	0.1297	0.0310	0.1201	0.0009	-0.0003	0.1118	0.0864	0.0805
MSE ↓	14.5351	14.9630	16.2005	420.4435	1637.1846	13.6581	13.0548	16.8056
RMSE ↓	3.7918	3.8331	3.9820	20.4440	40.3374	3.6549	3.5714	4.0536
Latency (ms) ↓	12.66 ms	349.46 ms	60.57 ms	6.61 ms	8.51 ms	892.44 ms	135.01 ms	489.20 ms
Throughput (FPS) ↑	79.0 FPS	2.9 FPS	16.5 FPS	151.2 FPS	117.6 FPS	1.1 FPS	7.4 FPS	2.0 FPS

* Notes: (↑) Higher is better | (↓) Lower is better | Bold: Optimal metric value across benchmarked models
Bicubic (Baseline): Standard interpolation baseline | Compact SRCNN RTL (1-16-8-1, 1,649 params, INT8 Q7) | SRCNN Original (1-64-32-1, 8,129 params, FP32)